

Can also invert rational $X(z)$ with long division 3

Ex. $X(z) = \frac{1 - z^{-1}}{1 - 1/2 z^{-1}}$

$|z| > 1/2$ $1 - 1/2 z^{-1}$ $\overline{) 1 - z^{-1}}$

$x[n] =$

0	$n < 0$
1	$n = 0$
$-1/2$	$n = 1$
$-1/4$	$n = 2$
$-1/8$	$n = 3$
$-1/16$	$n = 4$

Long division steps:

$$\begin{array}{r} 1 - z^{-1} \\ 1 - 1/2 z^{-1} \overline{) 1 - z^{-1}} \\ \underline{1 - 1/2 z^{-1}} \\ -1/2 z^{-1} + 1/4 z^{-2} \\ \phantom{1 - 1/2 z^{-1}} \underline{-1/2 z^{-1} + 1/4 z^{-2}} \\ \phantom{1 - 1/2 z^{-1}} \phantom{1 - 1/2 z^{-1}} -1/4 z^{-2} + 1/8 z^{-3} \\ \phantom{1 - 1/2 z^{-1}} \phantom{1 - 1/2 z^{-1}} \phantom{1 - 1/2 z^{-1}} \underline{-1/4 z^{-2} + 1/8 z^{-3}} \\ \phantom{1 - 1/2 z^{-1}} \phantom{1 - 1/2 z^{-1}} \phantom{1 - 1/2 z^{-1}} \phantom{1 - 1/2 z^{-1}} -1/8 z^{-3} + 1/16 z^{-4} \\ \phantom{1 - 1/2 z^{-1}} \phantom{1 - 1/2 z^{-1}} \phantom{1 - 1/2 z^{-1}} \phantom{1 - 1/2 z^{-1}} \underline{-1/8 z^{-3} + 1/16 z^{-4}} \\ \phantom{1 - 1/2 z^{-1}} \phantom{1 - 1/2 z^{-1}} \phantom{1 - 1/2 z^{-1}} \phantom{1 - 1/2 z^{-1}} \phantom{1 - 1/2 z^{-1}} -1/16 z^{-4} \end{array}$$

why we have to apply this rules and also when

What you are seeing are **rules for choosing the correct inverse Z-transform**, and they are applied **because a rational $X(z)$ does not uniquely determine $x[n]$ unless the Region of Convergence (ROC) is specified**.

I will explain **why the rules exist** and **when to apply each one**, using your example.

1. Why do we need these rules?

A rational Z-transform

$$X(z) = \frac{1 - z^{-1}}{1 - \frac{1}{2}z^{-1}}$$

can correspond to **different time-domain signals** depending on the **ROC**.

Reason:

- The Z-transform is **many-to-one**
- Poles alone do **not** fix causality or sidedness
- The **ROC determines whether the sequence is right-sided, left-sided, or two-sided**

Therefore, rules are required to select the **physically correct expansion**.

2. Key rule being applied in your notes

Rule:

If the ROC is outside the outermost pole ($|z| > |p|$), the sequence is right-sided (causal).

In your example:

- Pole at $z = \frac{1}{2}$
- ROC given: $|z| > \frac{1}{2}$

✓ Hence, the signal is **right-sided (causal)**

✓ So we must expand $X(z)$ in **negative powers of z**

This is why **long division / power series expansion** is used.

3. Why long division is used here

For causal signals:

$$X(z) = \sum_{n=0}^{\infty} x[n]z^{-n}$$

To find $x[n]$, we must rewrite $X(z)$ in this form.

That is exactly what long division does:

$$\frac{1 - z^{-1}}{1 - \frac{1}{2}z^{-1}} = 1 - \frac{1}{2}z^{-1} - \frac{1}{4}z^{-2} - \frac{1}{8}z^{-3} - \dots$$

From this:

$$x[0] = 1, \quad x[n] = -\left(\frac{1}{2}\right)^n, \quad n \geq 1$$

And:

$$x[n] = 0, \quad n < 0$$

4. When do we apply these rules?

Case 1: $|z| > |p|$ (outside poles)

- Right-sided (causal)
- Use **power series in z^{-1}**
- Use **long division or geometric expansion**

✓ This is your case.

Case 2: $|z| < |p|$ (inside poles)

- Left-sided (anti-causal)
 - Expand in **positive powers of z**
 - Long division is still used, but reversed
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Case 3: ROC not given

- Inverse Z-transform is ambiguous
 - You must state **multiple possible $x[n]$**
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5. Why not partial fractions here?

Partial fractions work best when:

- Simple poles
- ROC already implies causality clearly
- You recognize standard Z-transform pairs

Long division is preferred when:

- ROC is explicitly given
 - You want **direct coefficient extraction**
 - Exam problems require showing causality clearly
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6. Summary (exam-ready)

- Rules exist because Z-transform inversion is not unique
- ROC determines signal sidedness
- $|z| > |p| \Rightarrow \text{causal} \Rightarrow \text{expand in } z^{-1}$
- Long division is applied to obtain the power series

- Coefficients of z^{-n} give $x[n]$

If you want, I can also:

- Show the **anti-causal case**
- Compare with **partial fractions**
- Convert this into **short exam notes**
- Explain how to decide instantly in exams

Just tell me which one you want.



it is not clear how do i understand that I have to do partial expansion

This confusion is very common. The key point is that **you do NOT “guess” partial fraction expansion.**

You **decide it from the ROC and the algebraic form.** Below is a **clean decision procedure** you can follow every time.

Step-by-step rule (this is the part you are missing)

Step 1: Look at the form of $X(z)$

$$X(z) = \frac{1 - z^{-1}}{1 - \frac{1}{2}z^{-1}}$$

This is a **proper rational function in z^{-1}**
(degree of numerator $\leq$ degree of denominator)

That already tells you:

- No polynomial division in z
- Only **series expansion or partial fractions**

Step 2: Find the pole and the ROC

Denominator:

$$1 - \frac{1}{2}z^{-1} = 0 \Rightarrow z = \frac{1}{2}$$

Given:

$$|z| > \frac{1}{2}$$

This is **outside the pole**.

Step 3: Apply the single most important rule

ROC outside the pole $\Rightarrow$ right-sided (causal) signal

Right-sided signals **must be expressible as**

$$X(z) = \sum_{n=0}^{\infty} x[n]z^{-n}$$

So you **must** rewrite $X(z)$ as a power series in z^{-1} .

Step 4: Now decide the METHOD

Ask **one simple question**:

Can I directly match a known Z-transform pair?

Known pair:

$$\frac{1}{1 - az^{-1}} \iff a^n u[n]$$

Your denominator matches this form.

But your numerator is **not a constant**, it is:

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   ...

So you **must split the fraction**.

This is where partial fraction / algebraic expansion comes from

You rewrite:

$$X(z) = \frac{1}{1 - \frac{1}{2}z^{-1}} - \frac{z^{-1}}{1 - \frac{1}{2}z^{-1}}$$

This is **not optional** — it is forced, because:

- Each term now matches a known transform pair
- Each term can be expanded in z^{-1}

Step 5: Expand each term (because ROC demands it)

Since $|z| > \frac{1}{2}$:

$$\begin{aligned}\frac{1}{1 - \frac{1}{2}z^{-1}} &= \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n z^{-n} \\ \frac{z^{-1}}{1 - \frac{1}{2}z^{-1}} &= \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^{n-1} z^{-n}\end{aligned}$$

Subtracting gives:

$$X(z) = 1 - \frac{1}{2}z^{-1} - \frac{1}{4}z^{-2} - \frac{1}{8}z^{-3} - \dots$$

Final extracted signal

$$x[n] = \begin{cases} 1, & n = 0 \\ -\left(\frac{1}{2}\right)^n, & n \geq 1 \\ 0, & n < 0 \end{cases}$$

The ONE rule to remember (exam gold)

You decide **partial expansion / series expansion** using this logic:

1. Find poles
2. Read the ROC

3. ROC outside poles $\Rightarrow$ causal
4. Causal $\Rightarrow$ expand in z^{-1}
5. If numerator blocks direct matching $\Rightarrow$ split the fraction

There is **no guessing** involved.

One-line exam decision rule

If ROC is outside the pole and $X(z)$ is rational in z^{-1} , rewrite it as a sum of terms of the form $\frac{z^{-k}}{1-az^{-1}}$ and expand.

If you want, I can now:

- Show a **left-sided** example
- Give you a **2-line decision flowchart**
- Show **when partial fractions alone are enough**

Tell me which one.



now show example of left sided and both sided

Below are **clean, minimal, exam-oriented examples** showing **left-sided** and **two-sided** inverse Z-transforms, using the **same pole** so the contrast is obvious.

1. Left-sided (anti-causal) example

Given

$$X(z) = \frac{1}{1 - \frac{1}{2}z^{-1}}, \quad \text{ROC: } |z| < \frac{1}{2}$$

Step 1: Pole and ROC

- Pole at $z = \frac{1}{2}$
 - ROC is **inside** the pole
-

Step 2: Apply the rule

ROC inside the pole $\Rightarrow$ left-sided signal

Left-sided signals must be expanded in **positive powers of z** .

Step 3: Rewrite for positive powers

$$\frac{1}{1 - \frac{1}{2}z^{-1}} = \frac{z}{z - \frac{1}{2}}$$

Now write:

$$\frac{z}{z - \frac{1}{2}} = -\frac{1}{\frac{1}{2}} \cdot \frac{1}{1 - \frac{z}{\frac{1}{2}}}$$

Step 4: Series expansion (valid for $|z| < \frac{1}{2}$)

$$X(z) = -\sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^{-n} z^n$$

Step 5: Time-domain signal

$$x[n] = -\left(\frac{1}{2}\right)^n u[-n - 1]$$

Final result (left-sided)

$$x[n] = \begin{cases} -\left(\frac{1}{2}\right)^n, & n \leq -1 \\ 0, & n \geq 0 \end{cases}$$

2. Two-sided example

Given

$$X(z) = \frac{1}{(1 - \frac{1}{2}z^{-1})(1 - 2z^{-1})}$$

$$\text{ROC: } \frac{1}{2} < |z| < 2$$

Step 1: Poles and ROC

- Poles at $z = \frac{1}{2}$ and $z = 2$
 - ROC is **between poles**
-

Step 2: Apply the rule

ROC between poles $\Rightarrow$ two-sided signal

This means:

- One part must be **right-sided**
 - One part must be **left-sided**
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Step 3: Partial fraction expansion

$$X(z) = \frac{A}{1 - \frac{1}{2}z^{-1}} + \frac{B}{1 - 2z^{-1}}$$

Solving:

$$A = \frac{1}{1 - 4} = -\frac{1}{3}, \quad B = \frac{1}{1 - \frac{1}{4}} = \frac{4}{3}$$

So:

$$X(z) = -\frac{1}{3} \frac{1}{1 - \frac{1}{2}z^{-1}} + \frac{4}{3} \frac{1}{1 - 2z^{-1}}$$

Step 4: Match ROC to each term

Term	Pole	ROC relative	noting
$\frac{1}{1-\frac{1}{2}z^{-1}}$	$\frac{1}{2}$	outside	right-sided
$\frac{1}{1-2z^{-1}}$	2	inside	left-sided

Step 5: Inverse Z-transform

$$x[n] = -\frac{1}{3} \left(\frac{1}{2}\right)^n u[n] + \frac{4}{3}(2)^n u[-n-1]$$

Final result (two-sided)

$$x[n] = \begin{cases} \frac{4}{3}(2)^n, & n \leq -1 \\ -\frac{1}{3} \left(\frac{1}{2}\right)^n, & n \geq 0 \end{cases}$$

3. One-page decision table (memorize this)

ROC	Signal type	Expansion
(	z	>
(	z	<
Between poles	Two-sided	Split terms

Key takeaway (this removes all confusion)

- ROC decides sidedness
- Sidedness decides expansion
- Expansion decides method

There is no ambiguity once ROC is known.

If you want, I can now:

- Give a 1-minute exam flowchart
- Show common mistakes examiners penalize
- Practice mixed questions

Tell me what you want next.



make a table for all cases when and how to do which one

Below is a **single, complete decision table** you can use in **any inverse Z-transform problem**.

This table answers **when** and **how** to choose **partial fractions, long division, or series expansion** — no guessing.

Inverse Z-Transform: Complete Decision Table

Step	Condition you check	What it means	What you MUST do	Method
1	$X(z)$ is rational	Algebraic inversion possible	Continue	—
2	Find poles	Roots of denominator	Needed for ROC decision	—
3	ROC given?	Inversion is unique	Use ROC	—
4	ROC: $(r_1 < z < r_2)$	$ z > \max r_i $	$ z > \max r_i $	$ z > \max r_i $
5	ROC: $(z < r)$	$ z < \min r_i $	$ z < \min r_i $	$ z < \min r_i $
6	ROC between poles	Two-sided	Split terms by poles	Partial fractions

Step	Condition you check	What it means	What you MUST do	Method
7	Numerator blocks direct match	Not a standard pair	Decompose numerator	Partial fractions
8	$\text{Degree}(\text{num}) \geq \text{Degree}(\text{den})$	Improper fraction	Polynomial division first	Long division
9	Simple pole form $1/(1 - az^{-1})$	Standard pair	Match directly	Table lookup
10	Mixed poles	Different sidedness	Assign per term	Partial fractions

Method Selection Summary (EXAM CORE)

Situation	Use this
ROC outside poles	Expand in z^{-1}
ROC inside poles	Expand in z
ROC between poles	Partial fractions + mixed expansion
Numerator $\neq$ constant	Split numerator
Improper rational	Polynomial division
Single clean pole	Direct pair

What each method gives you

Method	Purpose
Partial fractions	Separate poles
Long division	Get power series

Method	Purpose
Series expansion	Extract $x[n]$
Direct pair	Fastest

Minimal exam flow (memorize this)

1. Find poles
2. Read ROC
3. Decide sidedness
4. Choose expansion direction
5. Decompose if needed
6. Write $x[n]$

Typical examiner traps

Mistake	Why it's wrong
Ignoring ROC	Inverse not unique
Wrong power expansion	Violates ROC
No sidedness stated	Incomplete answer
Mixing expansions	Mathematically invalid

One-line golden rule

ROC → sidedness → expansion → method

If you want, I can also:

- Convert this into a **one-page PDF cheat sheet**
- Give **10 solved mixed practice problems**
- Show **how marks are allocated in exams**

Tell me what you want next.

